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Revealing Novel Connections Between Space Weather and the Power Grid: Network Analysis of Ground-Based Magnetometer and Geomagnetically Induced Currents (GIC) Measurements

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Key Points:

- The connections between space weather and Geomagnetically Induced Currents are illuminated using network analysis on a current data set released by power grid utilities
- Deviations from average currents are 1.83 times more likely when there are deviations from average magnetic behavior
- The magnetometer deviation most indicative of a current deviation is often not the closest to the current measurement site

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Abstract The growing depth and breadth of data spanning the solar-terrestrial environment requires new ways of representing and analyzing the available information. This paper applies one such new data representation—network analysis—to the study of Geomagnetically Induced Currents (GICs) in electric power lines. This work uses newly available electric current data collected by power utilities through the the Electric Power Research Institute (EPRI) SUNBURST project and magnetometer data from the Super Magnetometer Initiative. The magnetometer data are analyzed using wavelet analysis. This new analysis method shows deviations to be more likely for equatorial stations close to water, which may be caused by the coast effect. The deviation likelihood is a complex function of latitude and magnetic local time. The GIC data are analyzed using "Quiet Day Curves" (QDCs) which help isolate geomagnetic disturbances. We find that current deviations are more common in the early morning sector, but this trend differs from station to station. These current and magnetometer data are represented in a network as nodes which are connected when both the current and magnetic measurements have a statistically significant deviation from their baseline behavior. This network is used to study the link between space weather and GICs. To do this, times when a current deviation exists are compared to times when magnetic deviations exist for each magnetometer - current sensor pair. Current deviations are, on average, 1.83 times more likely when there are magnetic deviations. However, some magnetometer deviations are more indicative than others, with the strongest probability multipliers reaching 3.

Plain Language Summary Geomagnetically Induced Currents (GICs) are harmful effects of space weather that have demonstrated their ability to damage power transformers and disrupt the electrical power grid, yet GIC data are rarely available to space weather researchers. A unique partnership with power grid utilities has led to a new data set of GIC observations for space weather research. This work combines GIC and magnetometer data from a number of stations around the globe to assess how often space weather is coincident with deviations in GICs, and which magnetometers are the best indicators of these deviations.

1. Introduction

Particles and energy from the sun can travel through interplanetary space and produce myriad impacts on Earth, the near-Earth space environment, and susceptible technology. Collectively, these impacts are known as "space weather," and they can have dramatic consequences for our technologically dependant society (Lanzerotti, 2001; Schrijver et al., 2015). Among the most important, yet ironically, least well specified impacts is to the electric power grid (D. H. Boteler, 2019; Pulkkinen et al., 2017). The most notable example of the power grid's susceptibility to space weather occurred on 13 March 1989 when anomalous Geomagnetically Induced Current (GIC) flows at the Hydro-Quebec power grid damaged equipment which caused a 9 hr blackout.

When enhanced space weather activity (e.g., the launch of a coronal mass ejection or a high speed solar wind stream (HSS) from the Sun) interacts with the Earth's magnetosphere, it produces a chain of complex interconnected physical processes, such as intense electric currents in the Earth's charged upper atmosphere, the

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ionosphere. Ionospheric currents induce electric currents along long conducting wires on the surface of the Earth, an important example of which is electrical transmission lines. The phenomenon of space weather-induced electric currents in the power grid is known as GICs (Pirjola, 2000; Viljanen & Pirjola, 1994). The challenge of understanding, forecasting, and mitigating GICs is a pressing one for the space weather community because they can disrupt the operation of the power system by overheating power transformers and generating excessive harmonics potentially resulting in the loss of power system equipment (Pulkkinen et al., 2017).

The study of GICs has been driven by widely available magnetometer data from, for example, the Super Magnetometer Initiative (SuperMAG (Gjerloev, 2009)) to quantify the geomagnetic disturbance due to space weather unified with a model of the Earth conductivity to produce the geoelectric field. The geoelectric field can then be used with a representation of the electrical power grid to calculate induced currents or GICs (Pirjola, 2002; D. Boteler & Pirjola, 2017). Despite the challenges and limitations that come from using assumptions of earth conductivity and plane wave propagation, these approaches have shown reasonable agreement with data. A unique partnership with power grid utilities via the Electric Power Research Institute (EPRI) SUNBURST Project (Lesher et al., 1994; EPRI, 2018a, 2018b) is leveraged to build off this analysis. The SUNBURST project is designed to collect high-quality, readily accessible data related to GICs associated with Geo-Magnetic Disturbances (GMDs). SUNBURST, operating since 1990, provides a sophisticated detection and recording network consisting of more than 50 transformer monitors at substations throughout North America. Current utilization of power utilities' data seldom goes beyond the immediate power network, and their union with relevant space weather data (e.g., solar wind parameters) is a rarity. This work uses direct observations of GICs collected at nine GIC sensor nodes throughout the United States over the course of 2018 to advance the study of GIC characteristics and their connection to widely available magnetometer data.

This work uses network analysis to link the GIC and magnetometer data. Network analysis (Boccaletti et al., 2006) has been a valuable tool in many fields of research, originating in the social sciences (Milgram, 1967), and finding more recent application in numerous disciplines such as biology, engineering, and geophysics (Donges et al., 2009; Malik et al., 2011; Steinhaeuser et al., 2011; Tsonis et al., 2006). Machine learning, of which network analysis is a subdivision, has been used in the space sciences before for example, (Camporeale, 2019). The efficacy of network analysis for discovery in space weather has been demonstrated (Dods et al., 2015, 2017; McGranaghan et al., 2017; Orr, Chapman, & Beggan, 2021; Orr, Chapman, Gjerloev, & Guo, 2021; Orr et al., 2019), and motivates its seminal use with GIC data in this work. Here, the nodes of the network are defined by GIC sensors and ground-based magnetometers. A GIC sensor is defined to be connected to a magnetometer at a moment in time if a current deviation is coincident with a magnetic deviation. Analyzing times of current deviations, magnetic deviations, and connections for the year of 2018 produces insight into the connections between magnetometers and GICs and moves toward a network approach for now-casting GIC risk from magnetometer measurements.

This paper is organized as follows: First, the GIC data is analyzed without the magnetometer data in Section 2. After using Quiet Day Curves (QDCs) to remove persistent trends and define a significant deviation, the likelihood of these deviations are studied as functions of Magnetic Local Time, time of year, and location. Second, the magnetic data is analyzed without the GIC data in Section 3. Wavelet analysis is used to baseline the data and define a significant deviation. The likelihood of these deviations is studied as a function of MLT and location. Third, the two datasets are joined and the network is analyzed in Section 4. A probability multiplier is used to describe the increase in the likelihood of a current deviation if a magnetic deviation is observed. The strength and location of strong probability multipliers is studied. Finally, summaries and conclusions are made in Section 6.

2. GIC Data

Data provided through the EPRI SUNBURST project by the Tennessee Valley Authority gives the transformer neutral Direct Current (DC) as a function of time for the year 2018 for nine stations. Each station is identified with a unique two-letter code to allow easy identification but to preserve anonymity for the stations. For all stations, the current is sampled at 1 Hz. This current is driven by a number of factors such as severe terrestrial weather, system operations such as maintenance and switching components, and space weather. A histogram of the current at each station is shown in Figure 1. Although the current stays between −0.5 and 0.5 Amps most of the time, there are a handful of very strong currents in this data set, particularly at stations Mo and Su. The shape of the distribution varies considerably from station to station.

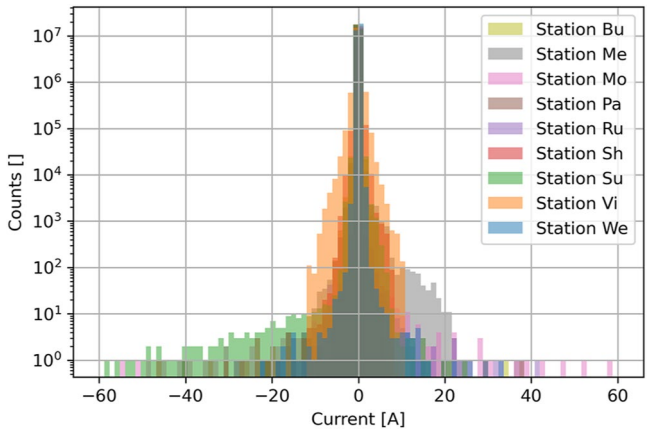


Figure 1. Histogram of measured Geomagnetically Induced Current at the nine stations used in this study.

There are small but persistent space weather signatures from the S_q current system (Yamazaki & Maute, 2017) in the data which make it difficult to set thresholds to define a deviation of the current from what might be considered background behavior. As an example, a GIC which occurs in an high-current region of the daily cycle will appear larger than one which occurs in a low-current region. In order to remove the persistent trends in the GIC data, in a uniform manner (accounting for differences experienced due to location and station characteristics), the data are grouped over 30-min intervals for each day, incorporating data from the same time of day, and from 5 days prior to and after the current day making 11-day of information total. These groups contain 30 min × 11 days = 5.5 hr of data. The mean and standard deviation (σ_{QDC}) of these data are computed and referred to as Quiet-Day Curves (QDC) of the expected current in the absence of magnetic activity. Data taken during times of active space weather are not considered. This technique is used for each available GIC measurement station to produce an estimate of the current due to the S_Q current system. This information is used to baseline the GIC data set, remove the S_Q signature, and distinguish between persistent daily trends and disturbance GIC variations. This technique is explained in further detail in (Kellerman et al., 2021).

Although the GIC data set stretches for all of 2018, a geomagnetically quiet period of the first five days of September 2018 are shown in Figure 2 to better show the daily variations. During this time period the D_{ST} varies between −21 and 18 nT. Since each station is unique, the y limits are also unique for each station. The blue line is the actual current measured, and the solid orange line is the QDC mean. Dashed orange lines show the 2σ bounds from the QDC. Note that the enforced y limits of $\pm\ 3\sigma$ hide some of the larger currents. There is not continuous data from all the transformers, and there also are not continuous QDCs even when there is data. For example, QDC fits are not available for Station Vi in this time period.

We define a current deviation as a time when the current differs from the QDC value by more two standard deviations from the mean. Put as an equation:

$$I \text{ is a deviation if } \frac{|I - QDC|}{\sigma_{QDC}} > 2 \tag{1}$$

This measure definitely responds to non-geomagnetic effects such as instrument maintenance or on/off changes, however since there are no markers in the data for such events they cannot be removed. Therefore, any measure will contain signatures of non-space weather effects. It is assumed that these events represent noise in the accumulated statistics and do not manifest in the final results. This definition provides a robust, empirical, and site-specific qualitative measure of departure of the current from "normal" behavior. These current deviations are identified visually in Figure 2 as times where the blue line exceeds the dashed orange lines. A sample time series during one of the more severe storms in the data set is shown in Figure 3 for one of the sites.

In the upper panel, the blue line shows the current measured, the solid orange line shows the QDC, and the dashed lines are the 2σ bounds. The times of a deviation ($|I - \text{QDC}| > 2\sigma_{\text{QDC}}$) are shown in the middle panel with either a zero (no deviation) or a 1 (deviation). The lower panel shows two space weather indices: the planetary K index (K_P) on the left axis and the Disturbance Storm Time (D_{ST}) index on the right. There is enhanced current both near the beginning of the storm late in the day on the 25th, and coincident with the minimum D_{ST} early on the 26th.

Consider when these deviations occur both in time and in Magnetic Local Time (MLT) through the 2D histogram in Figure 4. The color in panel a shows the probability of a deviation at any of the sites as a function of time and MLT. For example, if the probability in a given MLT and date bin is 50%, that means that half of the stations were deviant on that day and at that MLT. Figure 4b shows the time-averaged probability as a function of just MLT. Current deviations are more probable between 5 and 10 MLT when averaged over the entire year for many of the stations such as We, Bu, and Mo, but not at other sites such as Ru and Sh. Deviations are also more probable at some times than others - as evidenced by vertical streaks in panel a in early June, late August, and early November. Some of these times are well aligned with Kp (shown in panel c).

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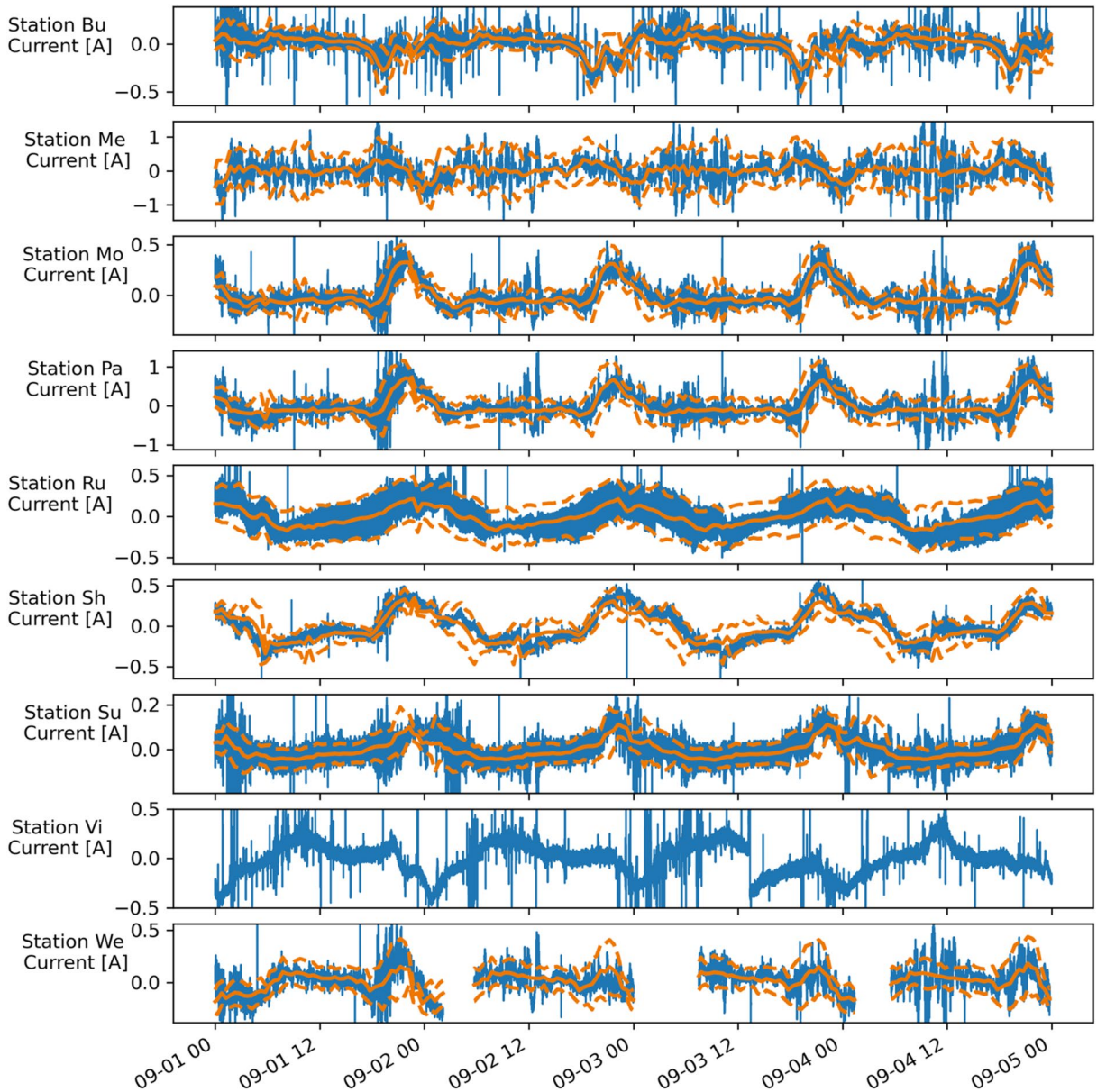


Figure 2. Geomagnetically Induced Current current as a function of time for all stations shown in blue. Solid orange lines show the quiet day curve while dashed lines show the $\pm 2\sigma$ bounds.

Next consider not when the current spikes occur, but where they occur. A map of all but one of the sites with the color and size indicating the deviation probability is shown in Figure 5. The map focuses on southeast continental United States to better show detail, which excludes the Vi station in California. Depending on the station, deviations occur between 3% and 15% of the time.

3. Magnetometer Data

Magnetometer data are provided by SuperMag which contains data from more than 200 magnetometers. For both the GIC data and the magnetometer data, all available data from 2018 is used. The data are processed to remove daily, yearly, and constant offset effects including the S_Q current system (Gjerloev, 2012). There are three steps to transform the magnetometer data and they are all shown in Figure 6 for the Boulder magnetometer.

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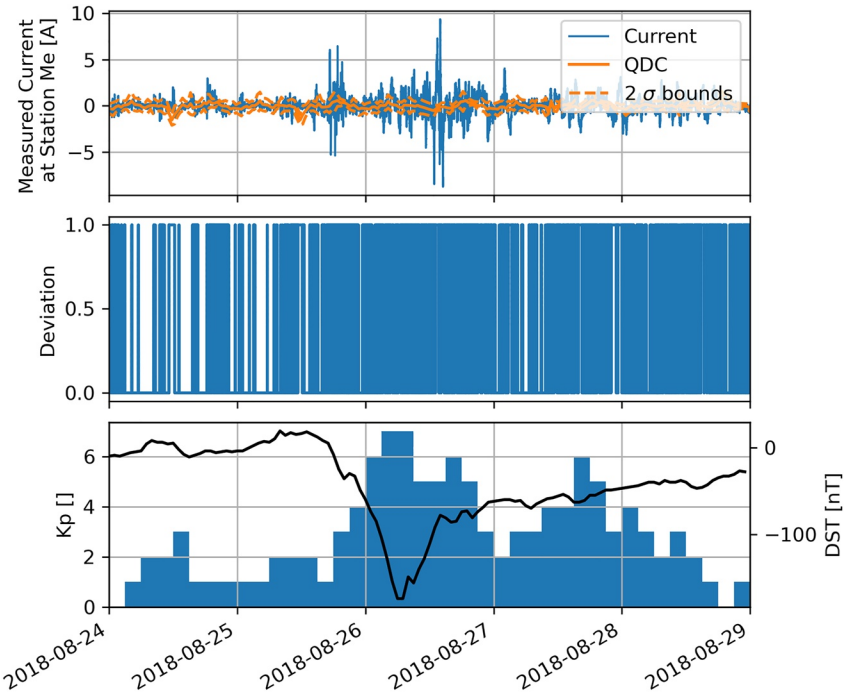


Figure 3. Time history of a Geomagnetically Induced Currents during a storm. The current, Quiet Day Curves, and bounds (top subplot). Index showing if the given measurement is a deviation (1) or not (0) (middle subplot). The D_{ST} (right) and Kp (left) indices (bottom subplot).

$$s = \frac{\|\boldsymbol{B}(t) - \boldsymbol{B}_m\|}{\|\boldsymbol{B}_m\|} \tag{2}$$

First, the magnetic field vector at a given station is converted to the scalar parameter, s as shown in Equation 2 where $\boldsymbol{B}_m$ is the median magnetic field vector for the whole year. In this analysis, only the horizontal magnetic field (north and east components) are considered since a change in those components induce horizontal electric fields (D. Boteler & Pirjola, 2017), but the results are insensitive to the use of the full field. This may be because the horizontal field is rarely perturbed without the vertical field also being perturbed. A time series plot of the full magnetic field is shown in Figure 6a for the same time time range used in Figure 3 showing the strongest storm in the data set, while the scalar parameter s is shown in Figure 6b.

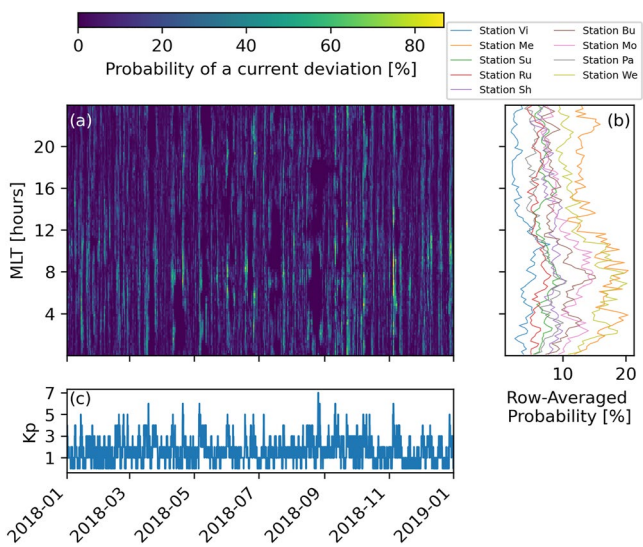


Figure 4. (a) Two dimensional histogram of current spike probability as a function of time and Magnetic Local Time (MLT). (b) Row-averaged probability histograms just as a function of MLT. (c) K_P as a function of time.

The second step is to perform a wavelet analysis on s and compute the significance level. Wavelet analysis have been used in the space sciences for many years to study magnetic phenomena for example, (Lui & Najmi, 1997; Nose et al., 1998; Ochadlick et al., 1993; Smith et al., 2020). The continuous wavelet transform with a sixth order Morlet wavelet is used to determine times and frequency of magnetic activity. Wavelet analysis shows the spectral content of a signal as a function of time. In some ways, it is like a set of Fourier transforms of the signal on different time windows. For this analysis, the pycwt python library (https://pypi.org/project/pycwt/) is used which is based on (Torrence & Compo, 1998). Figure 6c shows the power in s as a function of time and frequency. It shows a strong multi-day period signal necessary to make the envelope for the strong magnetic field perturbations that peak on the 26th. There is also a ~3 hr periodicity coincident with the main spike in the magnetic field and scalar parameter just before noon on the 26th. Another

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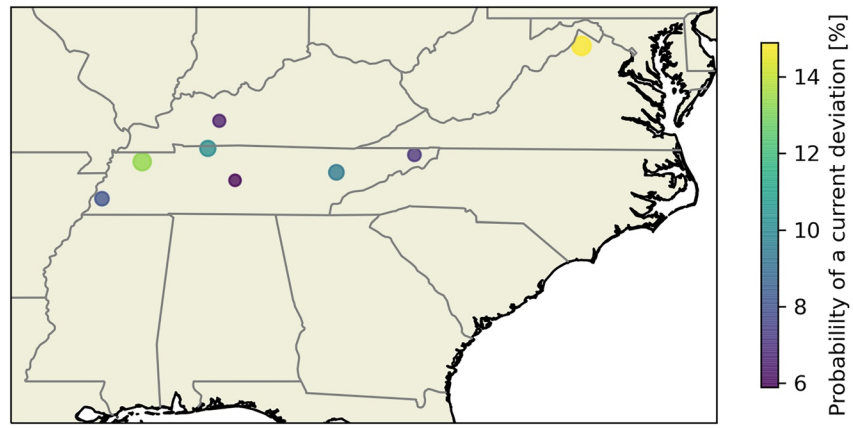


Figure 5. Deviation probability for all but one site. Only a small region of the southeastern Unites States (latitude between 30° and 40°, longitude between −92° and −75°) is shown to show detail which omits one west-coast station.

significant ∼3 hr periodic structure can be seen in the north and eastern $\boldsymbol{B}$ field components, the scalar parameter, and the wavelet analysis in the afternoon on the 27th.

This power is compared to an empirical red series to establish significance. The method for creating this red series if fully described in (Allen & Smith, 1996) and (Gilman et al., 1963) but we say here that it involves varying the auto-regression coefficient α so that s is well modeled by the series $s_{n+1} = \alpha s_n + z$ where z is a random variable with zero expected value. When α is very close to 1, the series has most of its power at low frequencies. When $\alpha = 0$, the series has equal power at all frequencies. For this example, $\alpha = 0.9985$ indicating that the series is very red.

A Chi-Squared test is used at the 95% confidence level to estimate the spectra of this red series. This is done using Equation 18 of (Torrence & Compo, 1998). This spectra is shown in Figure 6c2 which has the same y axis as Figure 6c. The thin red lines in Figure 6c encircle regions where the power is higher than in Figure 6c2. This means that the data inside the red lines is less than 5% likely to have occurred by chance. Using a red series (which has some temporal structure) to compute the significance level is a more selective criterion than just shuffling the time series.

Finally, the average significance ratio is computed as a function of time and shown in Figure 6d. This is done by dividing the power shown in Figure 6c by the power from the red series shown in Figure 6c2. This ratio is then averaged across periods from 3 s up to 3 days. When this ratio is above 1, we define that magnetometer to be experiencing a magnetic deviation. This definition classifies some of the 24th and 28th as being deviant despite fairly modest magnetic variations that might not be highlighted in other methods.

The fraction of time that each magnetometer was deviant is shown as color in Figure 7. Equatorial stations tend to have the most frequent deviations, and the most deviant of the equatorial stations tend to be close to water. This may be because of three reasons - first, equatorial sites are the most sensitive to the equatorial electrojet (EEJ) (Appleton, 1946; Yamazaki & Maute, 2017). However, the SuperMag processing technique may remove most of the EEJ signal from the data (Gjerloev, 2012). Second, in some areas proximity to water increases the variability of magnetic field data through the coast effect (Parkinson, 1959, 1962), where the vertical field perturbations is abnormally large and correlated to the onshore horizontal field. This effect is primarily driven by the difference in conductivity between the ground and the water. Lastly, magnetometers react strongly to ring current enhancements, which are strongest near the magnetic equator. The SuperMag network at equatorial latitudes has even been shown to react to auroral and cross-magnetotail currents (Newell & Gjerloev, 2012). These findings differ from prior work based not on deviations measured with wavelets, but instead on the strength of the horizontal electric field and magnetic field rate. These strength and electric field informed studies find stronger responses at high latitudes (Blake et al., 2021; Woodroffe et al., 2016). Our method for defining a deviation is statistically based and does not use a priori values for the magnetic field strength or rate to qualify a deviation. Future work could use a different frequency band to establish deviations, or explore using a priori values for the magnetic field strength and/or rate.

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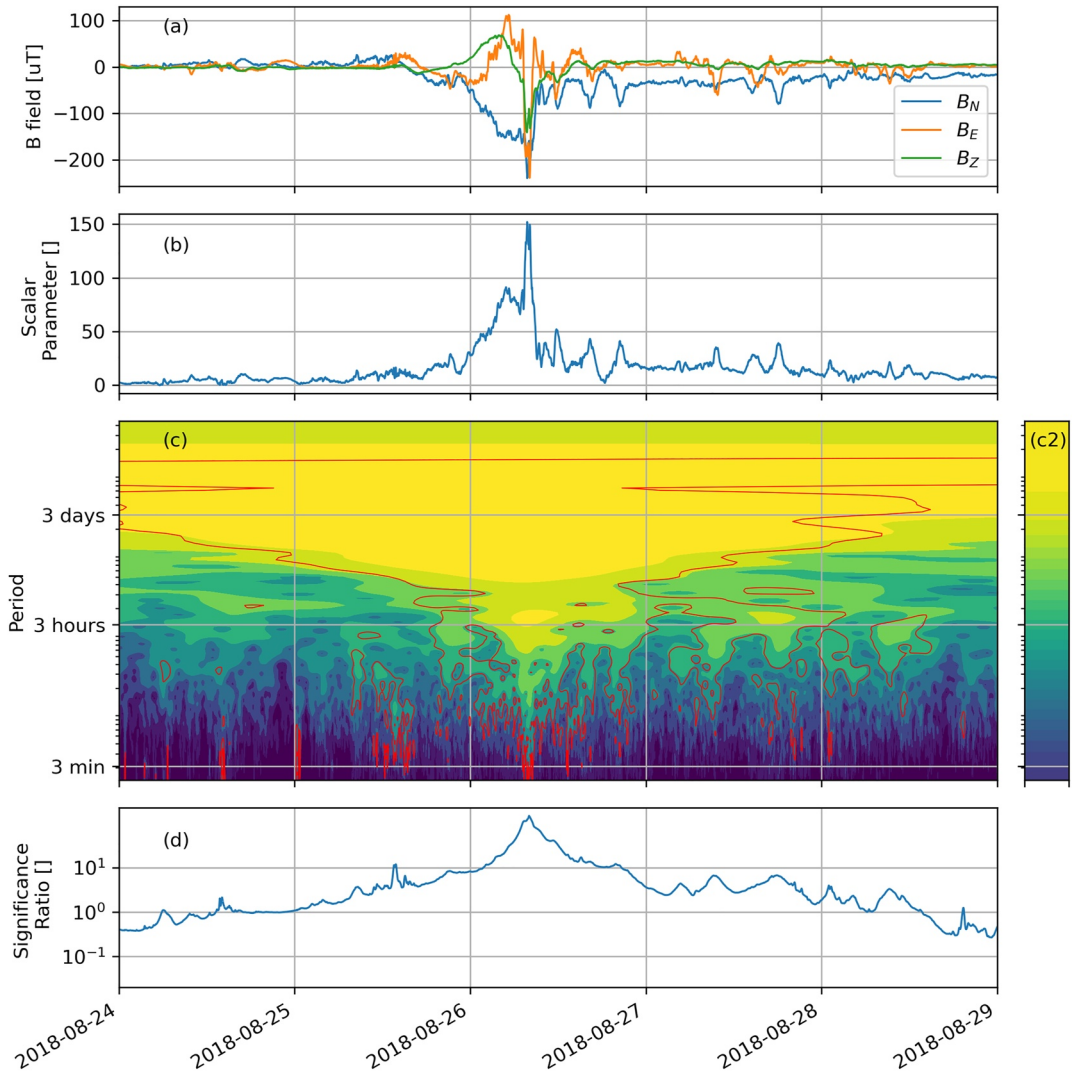


Figure 6. Example Wavelet analysis process: (a) Magnetic field at Boulder, CO magnetometer as a function of time. (b) Scalar parameter formed from magnetic field. (c) Wavelet periodogram for the scalar parameter. Color indicates power. (c2) Red noise spectra used to compute significance. (d) Significance ratio for periods between 3 s and 3 days.

The probability of magnetic deviations both for all data, and the data when K_P is greater than three, is plotted as a function of the magnetic latitude in Figure 8. A dashed line shows the median probability of magnetic activity in nine bins stretching from 0 to 90 degrees. The probability is highest at equatorial latitudes and slowly drops to a minimum near 70 degrees. Poleward of this, the probability of activity increases again. The $K_P > 3$ data differs from the rest of the data in that the percentage of the time in which the stations are active is much higher - nearly all the time for most magnetometers below 40 degrees absolute magnetic latitude. The transition near 70 degrees is also sharper.

Next, consider when in both time and MLT magnetic deviations occur. Another 2D histogram, very similar to Figure 4, of time and MLT is shown in Figure 9. There are a handful of strong magnetic events in this year as evidenced by the strong vertical yellow lines in Figure 9a. Many of these are well-aligned with K_P, shown in Figure 9c. The most prominent yellow line is centered around Aug 26, which also had the highest K_P value (7+) for all of 2018 and is the focus of Figures 3 and 6. Unlike the Tennessee Valley Authority data, the magnetic activity is spread across MLT for many of the storms because the SuperMAG stations are located across many longitudes.

The magnetic deviations are most probable in the late night and early morning which is where the majority of the magnetic activity is expected for example, (Newell & Gjerloev, 2014). Figure 9b shows a histogram of magnetic

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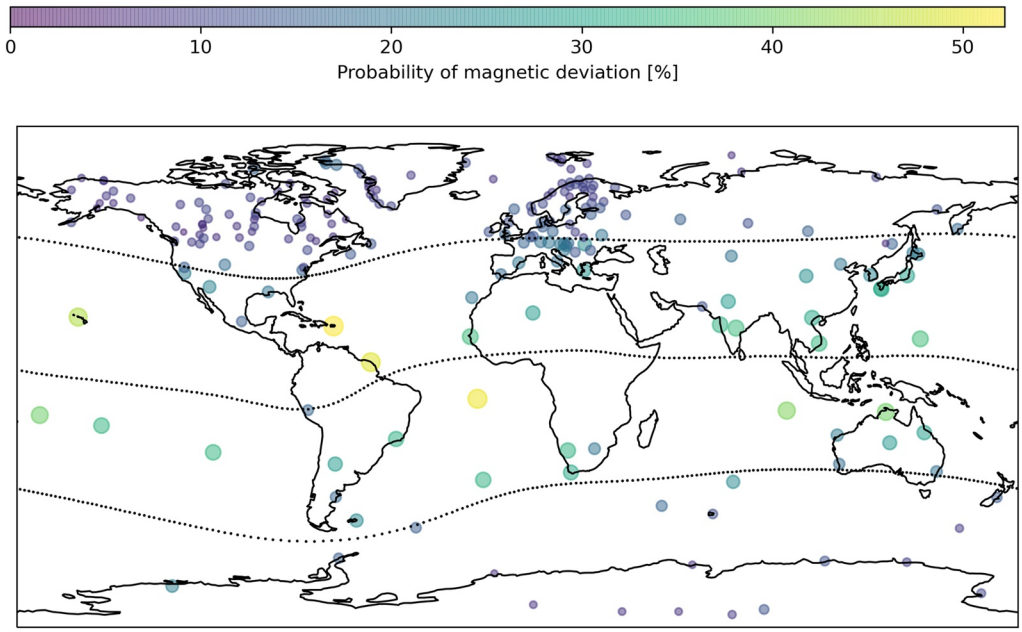


Figure 7. Map of probability of magnetic deviations for all magnetometers in SuperMAG network. Dotted lines show −45°, 0°, and 45° magnetic latitudes on the summer solstice, 2018.

deviations for each site with the magnetic latitude of the site shown as color. The average deviation probability of all the magnetometers as a function of MLT is shown as a dashed black line. This subplot shows two populations of magnetometers - those that have their peak deviation probabilities near magnetic midnight and those that have their peak deviation probability near magnetic noon. All but one of the noon-peaking stations is equatorial (dark blue in color) which points to the EEJ and equatorial fountain effect as an explanation for some of this bi-modality.

To further investigate this phenomenon, the ratio of the average magnetic deviation probability from 11 MLT to 13 MLT (noon centered) to the the average magnetic deviation probability from 23 MLT to 1 MLT (midnight centered) as a function of magnetic latitude is shown in Figure 10. When this ratio is 1, a magnetometer is as likely to be deviant near noon as near midnight. When it is larger than 1, it is more likely to deviant at noon. When it is less than 1, the magnetometer is more likely to deviant at midnight. The color and size of the points indicates the deviation probability for that site across all MLTs through the year. Confirming the trends observed in Figure 8, equatorial sites are much more active as evidenced by the large yellow points at the left end of the plot in Figure 10. There is also a clear decreasing trend in the noon to midnight ratio - indicating a preference for near-midnight magnetic deviations—until a magnetic latitude near 70 degrees. At these high latitudes, near-midnight deviations are more than 10 times more likely than near-noon deviations. After this point, the trend reverses and the stations are much more likely to be active near local noon. This trend is corroborated by (Ngwira et al., 2013), albeit with a slightly different transition magnetic latitude near 30–40 degrees. The lower transition to the cusp is likely due to Ngwira et al. only analyzing extreme events, when the auroral oval moves down to lower latitudes for example, (Carbary, 2005).

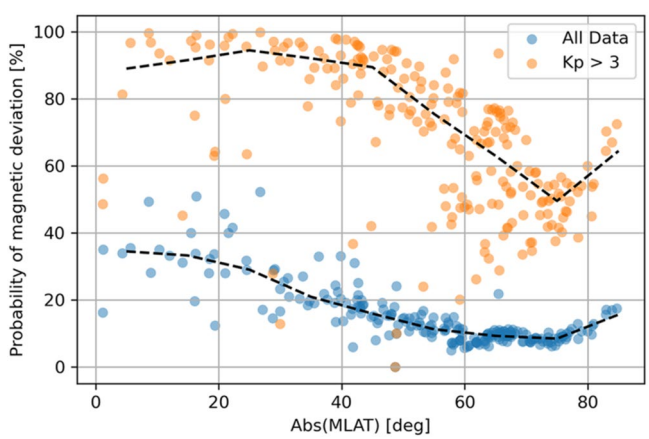


Figure 8. Probability of magnetic deviations as a function of magnetic latitude.

4. Statistical Connections

This section uses the definition of a current deviation in Equation 1 and the definition of a magnetic deviation (using wavelets) to construct a network. In this network, every magnetometer and every current measurement site is a node. Any magnetometer can have a connection to any current measurement

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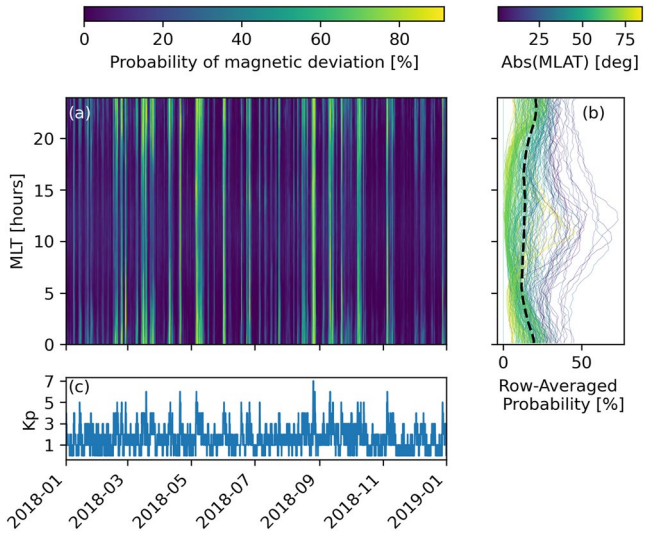


Figure 9. (a) Two dimensional histogram of magnetic deviation probability as a function of time and Magnetic Local Time (MLT). (b) Row-averaged probability histograms just as a function of MLT for each magnetometer station. Color indicates magnetic latitude (c) K_P as a function of time.

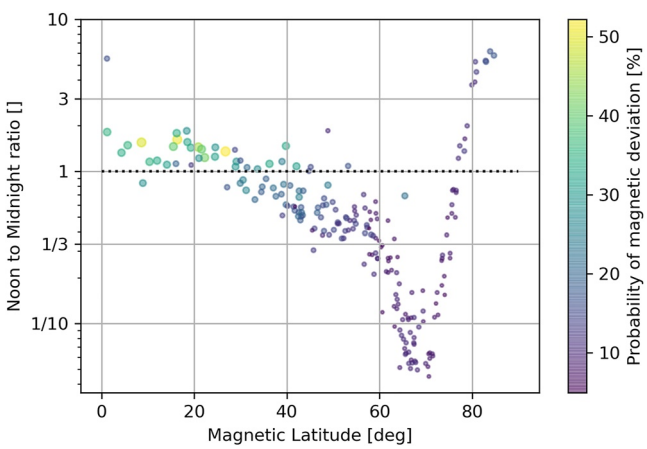


Figure 10. Ratio of noon (Magnetic Local Time (MLT) between 11 and 13) deviation probability to midnight (MLT between 23 and 1) deviation probability as a function of magnetic latitude.

site, but magnetometers (current measurement stations) cannot be connected to other magnetometers (current measurement stations). To determine if two nodes are connected, the amount of time during which there is a deviation at the magnetometer, a deviation at the current measurement site, deviations at both, or deviations at neither is used. The magnetometer data (1 min cadence) is linearly interpolated onto the GIC data timescale (1 s cadence) to allow for comparison. Figure 11 visualizes these times for four representative magnetometer-GIC sensor pairs. In each diagram, the purple circle represents all the times with magnetic deviations, and the tan circle represents all the times with current deviations. The intersection of the purple and tan circles (gray) shows times when both the current and magnetometer measurements are deviant. The large pink circle represents the entire year, and the area not occupied by any other circle represents the amount of time with no deviations at all. The number of seconds divided by 1000 for these subsets are also printed in the circles. Consider case (a); there are about 2.9 + 3.3 = 6.2 million instances of deviant currents and 7.1 + 3.3 = 10.4 million cases of magnetic deviation, 3.2 million of which are simultaneous. The total of all the numbers in each diagram gives the number of seconds in the year 2018.

Figure 11a represents a strong connection - magnetic and current deviations often occur simultaneously. Figure 11b shows a exemplary case of "false positives" where magnetic deviations are common, but current deviations are not. For this case, there are more than 14 million magnetic deviations, but only 1.5 million current deviations. Figure 11c shows an example of a false negative - where the magnetometer is not deviant even though there are many current deviations. In this case, the magnetic deviations only coincide with 643 thousand of the more than 5 million current deviations at this node. Lastly, Figure 11d shows a pair that anti-correlates, current deviations are less likely at this node if magnetic deviations are observed at the magnetometer. This analysis can be used to calculate the probability multiplier for a given current measurement station - magnetometer pair. The probability multiplier is the ratio of the probability of a current deviation given magnetic deviation to the unconditional probability of a current deviation. Using Bayes' law, this is given as:

$$PM = \frac{p(g|m)}{p(g)} = \frac{p(g \cap m)}{p(g)p(m)} \tag{3}$$

where $p(g)$ is the probability of current deviations, $p(m)$ is the probability of magnetic deviations, and $p(g \cap m)$ is the probability of coincident magnetic and current deviations. If all current deviations are independent from magnetic deviations, $p(g \cap m)$ would be equal to $p(g) \times p(m)$ and the probability multiplier would always be equal to 1. A probability multiplier greater than one indicates increased current deviation probability with magnetic deviations. When this quantity is greater than 1, we define the magnetometer - current measurement station to be connected in the network.

Denoting the number of occurrences in a year of magnetic deviation as N_M, the number of current deviations as N_G, the number of coincident deviations as N_B, and the total number of times (The number of seconds in 2018-31,536,000) as N_A, the probability multiplier is given by:

$$PM = \frac{N_A N_B}{N_G N_M} \tag{4}$$

As an example, the probability multiplier for the pair shown in Figure 11a is computed as $\frac{31,536 \times 3,280}{10,412 \times 6,229} \approx 1.6$. Recall that Figure 11 only shows exclusive probabilities so $N_G = 2,949 + 3,280 = 6,229$ and $N_M = 7,132 + 3,280 = 10,$

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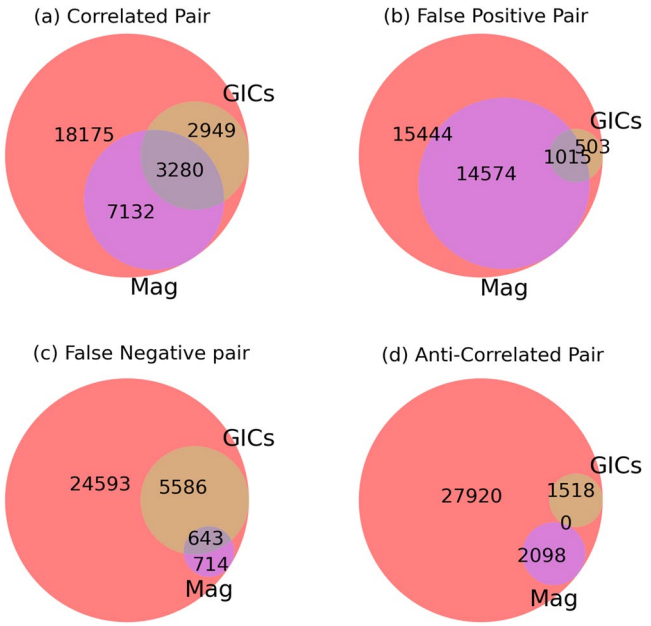


Figure 11. Venn diagrams for four representative magnetometer-Geomagnetically Induced Currents sensor pairs. In each diagram, the large pink circles represent all times. Tan circles represent times where significant DC transients are observed, purple circles represent times where magnetic activity is observed, and the gray intersection represents times where both magnetic activity and significant DC transients are observed. Numbers in each sector count the total times for each condition.

412. The probability multiplier for each GIC measurement node—magnetometer pair is computed and shown in Figure 12.

The color in the center plot shows how much more probable a current deviation at the station shown on the *y* axis is if magnetic deviation is observed at the magnetometer on on the *x* axis. For example, current deviations at Station Vi are more than three times more likely if magnetic deviations are observed in Kenai College, AK. Only the 10 most influential and 10 least influential, as determined by the probability multiplier, magnetometers are shown. The row and column average of this matrix are shown in the top and right bar plots. The column averages (top) show the average influence of a given magnetometer across all GIC measurement nodes. The most influential station (Kenai College) has an average probability multiplier near 2.5, and the least influential station (Trelew, Argentina) has an average multiplier of just under one. The row means (right) show the susceptibilities of a given station to all the magnetometers (not just the 20 shown). All the row means are significantly larger than one, meaning that all of the stations are (on average) affected by magnetic deviations. The bottom plot shows the probability of deviations at that magnetometer, and the left plot shows the probability of current deviations at the given station. These subplots show the same information as presented in Figures 7 and 5, respectively. The average probability multiplier for the entire matrix is 1.83 but for the best connected pairs it is higher than 3.

Many of the most influential magnetometers are located at high latitudes, and therefore it is reasonable to investigate if the observed links are due solely to heightened magnetic activity at high latitudes. This latitudinal effect is partially accounted for due to the empirical and magnetometer-specific processing technique. Furthermore, there is significant variation within high-latitude magnetometers. For example, Kenai College and Trapper Creek have different strength connections to the GIC sensors. Many other high-latitude stations are not ranked among the 10 most influential and do not appear on this plot. If high magnetic latitude alone were enough to explain the effect, then all 10 of the spots would be filled by high-latitude magnetometers and no high-latitude magnetometers would be excluded. The fact that magnetometers from a variety of magnetic latitudes have the highest probability multipliers is a testament to the significance of these sites over and above increasing activity with latitude.

Another approach to baseline the probability multipliers is to augment the magnetometer significance ratio time series and recompute the probability multipliers. This augmentation should recreate the mean behavior of the data, but have different temporal structure so that any connections that re-appear using this data can be ascribed to only the mean activity at those sites, and not due to any temporal structure or causal relationship. We explore three approaches for data augmentation:

1. Shuffle the original time series to make the augmented data. This method perfectly replicates the distribution (i.e., a histogram) of the original data, and the augmented data has no temporal structure. However, the augmented time series is guaranteed to be white (equal power at all frequencies) regardless of the spectra of the original time series, so it doesn't re-create all the features of the input data.
2. Roll the original time series by at least one month and no more than 11 months to make the augmented data. The range is imposed to ensure that the augmented series is sufficiently different from the original one. This approach also perfectly reproduces the distribution of the original series, and will additionally reproduce the spectra, ignoring edge effects.
3. This method follows (Dods et al., 2015), with some modifications for our unique data set. The original time series is linearly interpolated onto a regular time grid to account for data gaps. This continuous time series is Fourier transformed. The phase is shuffled, and then the data are inverse Fourier transformed. The magnitude of the new series is then used as augmented data. This approach recreates the spectra, but has a different distribution since the phase is not random in the original series.

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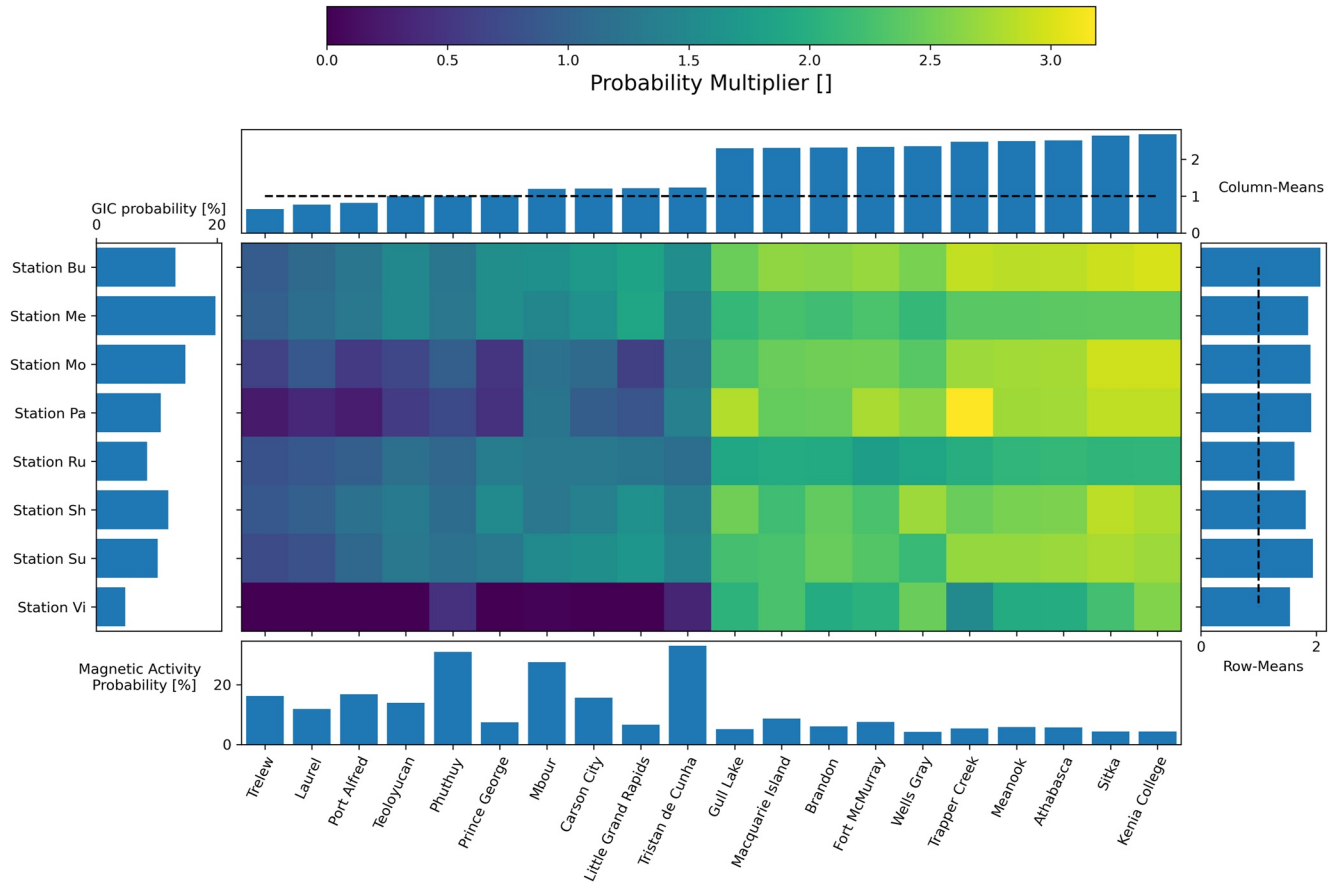


Figure 12. Probability multipliers for Geomagnetically Induced Currents node—magnetometer pairs with magnetic and current deviation means. See text for details.

These three augmentations differ in how they recreate the temporal structure of the data, but it is worth noting that the probability multipliers are not computed with any past or future information from the magnetometer time series. This reduces the importance of getting the temporal structure of the augmented data to match the original data perfectly. All classes of augmented data are used to compute a set of probability multipliers and some statistics are shown in Table 1 alongside the statistics for the original data.

The 10% and 90% quantiles are the values that are larger than 10% and 90% of the data, respectively. The range between these two numbers encompasses the middle 80% of the distribution. While the probability multipliers for the augmented data lie in a narrow range centered around 1.0, which is expected for independent processes, thb probability multipliers for the real data are significantly higher. For example, while the 90% quantile of the original data is 2.274, the largest 90% quantile for any of the augmented datasets is 1.239. All of the means and medians of the augmented data are close to 1, with the largest departure being by 0.015. In contrast, the real data has a mean and median of 1.852 and 1.832, respectively. This shows that there is information in the magnetic significance ratio beyond that which is inevitable due to heightened magnetic activity at high latitudes.

Table 1
Statistics for Probability Multipliers Made Using Original and Augmented Magnetic Significance Ratios

Data type	10% quantile	Median	Mean	90% quantile
Original Data	1.403	1.852	1.832	2.274
Shuffled Data	0.986	1.000	1.000	1.013
Rolled Data	0.789	0.985	1.005	1.239
Shuffled Phase	0.957	0.998	0.993	1.026

These connections are also visualized through a map. The average influence of each magnetometer (top subplot in Figure 12) is shown as color in Figure 13. The 30 strongest connections are shown as thin black lines. The strongest connections, as indicated by the probability multiplier, are between GIC nodes and high-latitude magnetometers in northern Canada and Alaska. In all of these connections, the magnetometer is to the West of the current measurement node. This may be due to the shape of the current flow in this region (Keiling et al., 2009). If this plot is re-made using the augmented

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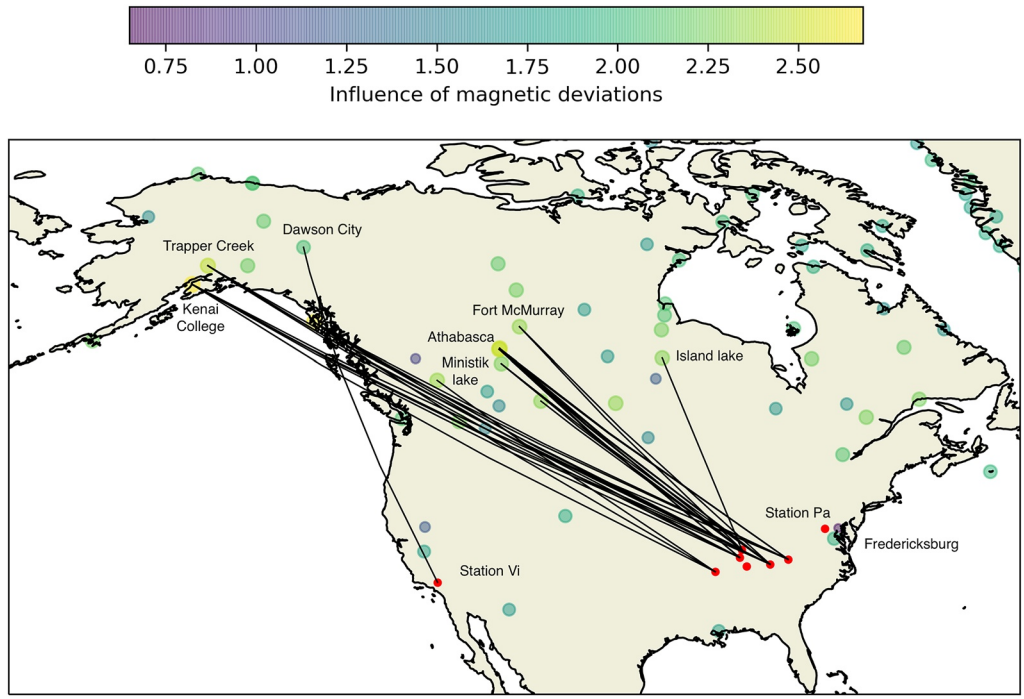


Figure 13. Map of strongest magnetometer-Geomagnetically Induced Currents node connections, color indicates average influence of magnetic deviation for all GIC measurement nodes.

probability multipliers, there is no trend in the connections. They are randomly distributed across the globe (not shown here).

5. Discussion

Previous work (EPRI, 2020) has shown that magnetic fields correlate with measured currents well when the separation is less than 300 miles, but this correlation diminishes as the distance increases out to 700 miles. In contrast, the strongest connection in this network is between Station Pa and Trapper Creek (in Alaska) with probability multiplier of 3.18. The Fredericksburg magnetometer is much nearer to Station Pa, but it has a lower probability multiplier of 2.07. Magnetic deviations at either of these magnetometers corresponds to higher current deviation probability at station Pa, but a deviation at Trapper Creek is about 50% more indicative of a current deviation despite being much further away. There are many possible reasons for this difference. First (EPRI, 2020) differs from this work in the definition of a current deviation using QDCs rather than correlation-based analysis. The use of QDC data makes it less likely that our analyses are affected by quiet variations than a correlation analysis of the original data. One of the important outcomes of this work is the use of the QDC data and is the first such data analysis using those data prepared by (Kellerman et al., 2021). Second, high-latitude magnetometers have a stronger response to space weather and therefore may have a higher signal to noise ratio to better predict current deviations.

To investigate this second explanation, consider Figure 14 shows two space weather indices in (a) to give geophysical context, and measurements from Fredericksburg and Trapper Creek in panels (b) and (c). The magnitude of the horizontal magnetic field rate is shown to follow (EPRI, 2017) in (d). It is given explicitly as:

$$\sqrt{\frac{dB_N}{dt}^2 + \frac{dB_E}{dt}^2} \tag{5}$$

for North and East components of the magnetic field B_N and B_E, respectively. Each magnetometer's significance ratio (power in the wavelet divided by the power in a random red spectrum) is shown in (e). Times when this ratio

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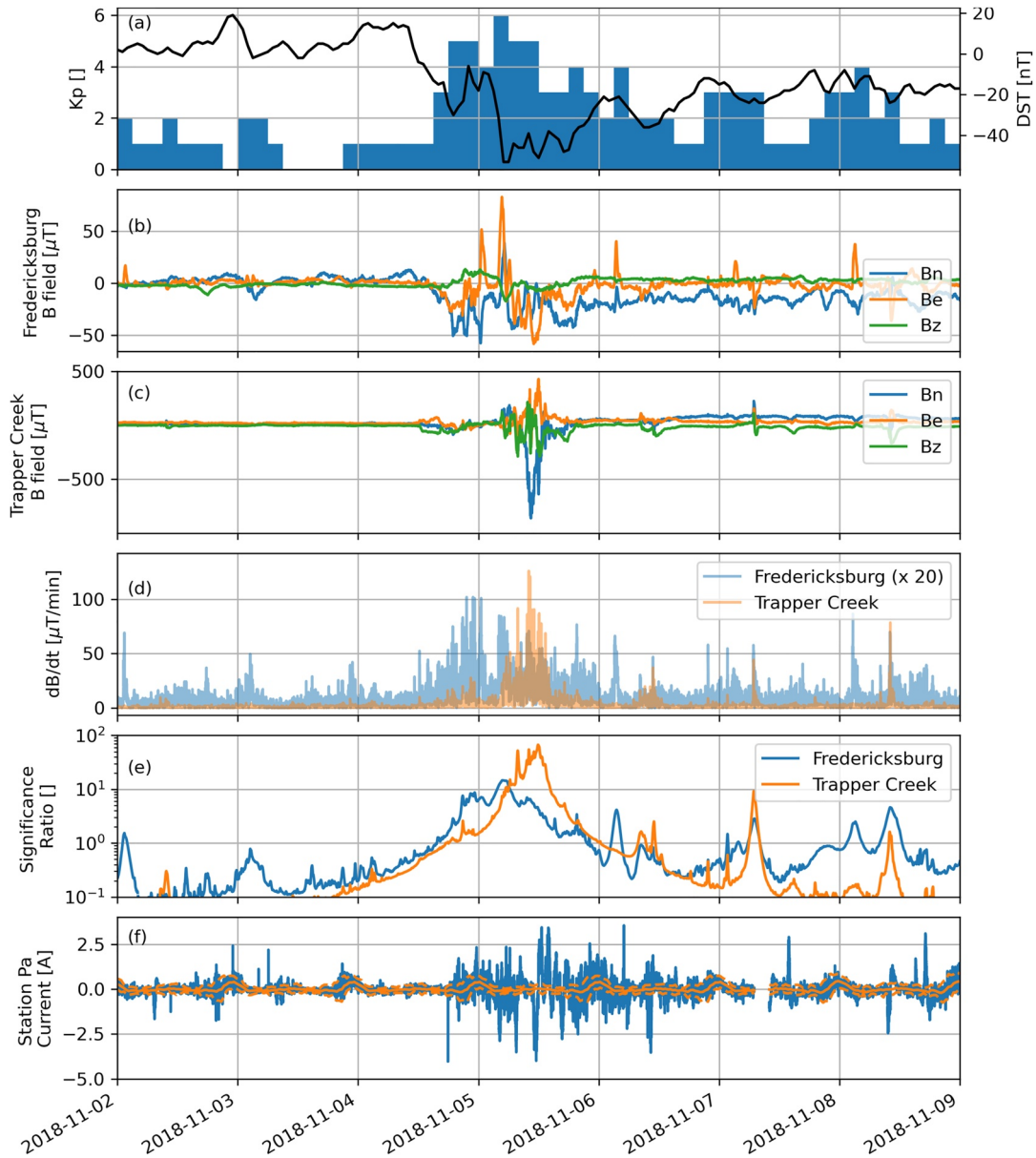


Figure 14. Time series for two magnetometers and Station Pa. (a) Disturbance Storm Time (D_{ST}) index (right) and Plantery K (K_p) index (left). (b) Magnetic field at Fredericksburg, close to the Geomagnetically Induced Currents measurement node. (c) Magnetic field at Trapper Creek, Alaska. (d) Horizontal magnetic field rate for both magnetometers. (e) Significance ratio of both magnetometers (f) Raw current with Quiet Day Curves and σ for Station Pa.

is larger than 1 are defined to be magnetic deviations. Lastly, the current measured at station Pa is shown in (f) with a solid orange line showing the QDC, and dashed orange lines showing the 2σ bounds.

The magnetic field at Trapper creek has much larger excursions relative to quiet time variability than Fredericksburg. This can be seen in panels (b–e), but it is easiest to see in panel (d) where a factor of 20 is used to give both magnetometers similar scale on the same axis. Even with this factor, the quiet time variability (for instance, on the first day) is much larger at Fredericksburg than Trapper Creek. Therefore, the effective signal to noise ratio is much higher at Trapper Creek than at Fredericksburg which may provide a clearer signal with which to predict current deviations.

We have provided the first exploration of the connections between magnetometers and GIC using network analysis and uncovered potential geophysical implications. We have discovered expected outcomes, such as the

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heightened probability multipliers of magnetometer activity at high latitudes and activity on GIC nodes, and less intuitive outcomes, such as a complex relationship between GIC activity and distance to the nearest magnetometer. We suggest that this lays the foundation for much future work to explore new geophysical insights from these network connections.

6. Conclusion

This paper offers a glimpse into what can be done with network analysis in the space sciences—particularly with the connection between space weather as evidenced by magnetometer measurements and deviations from the expected currents. The analysis of current data reveals a trend of higher probability current deviations in the early morning MLT sector. However, this is more dramatic at some stations than others. The magnetometer analysis showed that the equatorial and coastal magnetometers are the most likely to experience a deviation due to the combination of the coast effect, Equatorial ElectroJet, and ring current enhancements. The latitude also drives the ratio of noon-to-night activity with equatorial sites being more deviant near local noon and high latitude sites being more likely to be deviant near midnight. This trend is reversed near the auroral boundary. Probabilistic analysis shows which magnetometers are the most indicative of current deviations in the power grid, and where they are located. Once the magnetic and GIC data are joined through probability multipliers, three key points emerge from this analysis:

1. Current deviations are, on average, 1.83 times more likely to occur when there is a magnetic deviation.
2. This analysis indicates that magnetometers far from the GIC nodes can possibly provide a better indication of these current deviations.
3. Without any prior knowledge of the physical system, the network analysis technique presented in this paper is able to identify a strong connection between current deviations occurring at mid-latitude stations, and a set of auroral and sub-auroral magnetometers that, in all likelihood, reflect the occurrence of substorm-related ionospheric currents. While the physics of this connection are not yet fully identified, it is nevertheless clear that this analysis provides a means of estimating the likelihood of current deviations. As such, this represents a significant step forward in GIC forecasting.

Future work could continue the network analysis of the magnetometer data to study the temporal and spatial extent of storms and substorms. A valuable next step for this work would be to study which magnetometers are the best indicators of future current deviations, and a predictive model could be created. If this work was re-created with metrics more relevant to harm posed to the power grid, this predictive model could help alert power system operators to potential GIC risk and allow them to mitigate the effects. Additional current data from northern Canada and Alaska would enable further investigation into the trend reported here that significant transients preferentially connect with magnetometers in northern Canada and Alaska rather than with magnetometers nearby.

Data Availability Statement

The data provided for this study are part of the Sunburst network. We are currently not able to release the GIC data directly, due to agreements with EPRI and the power utilities who provided data for this analysis, however, we have posted the GIC-QDC data on https://zenodo.org/record/4444068#.YUoLl9NKj0o.

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References

Allen, M. R., & Smith, L. A. (1996). Monte Carlo SSA: Detecting irregular oscillations in the presence of colored noise. *Journal of Climate*, *9*(12), 3373–3404. https://doi.org/10.1175/1520-0442(1996)009<3373:mcsdio>2.0.co;2

Appleton, E. V. (1946). Two anomalies in the ionosphere. *Nature*, *157*, 691. https://doi.org/10.1038/157691a0

Blake, S. P., Pulkkinen, A., Schuck, P. W., Glocer, A., & Tóth, G. (2021). Estimating maximum extent of auroral equatorward boundary using historical and simulated surface magnetic field data. *Journal of Geophysical Research: Space Physics*, *126*(2), e2020JA028284. https://doi.org/10.1029/2020JA028284

Boccaletti, S., Latora, V., Moreno, Y., Chavez, M., & Hwang, D.-U. (2006). Complex networks: Structure and dynamics. *Physics Reports*, *424*(4), 175–308. https://doi.org/10.1016/j.physrep.2005.10.009

Boteler, D., & Pirjola, R. (2017). Modeling geomagnetically induced currents. *AGU Space Weather*, *15*, 258–276. https://doi.org/10.1002/2016SW001499

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Boteler, D. H. (2019). A 21st century view of the March 1989 magnetic storm. *Space Weather*, *17*, 1427–1441. https://doi.org/10.1029/2019SW002278

Camporeale, E. (2019). The challenge of machine learning in space weather: Nowcasting and forecasting. *Space Weather*, *17*(8), 1166–1207. https://doi.org/10.1029/2018SW002061

Carbary, J. F. (2005). A Kp-based model of auroral boundaries. *AGU Space Weather*, *3*(10), S10001. https://doi.org/10.1029/2005SW000162

Dods, J., Chapman, S., & Gjerloev, J. (2017). Characterizing the ionospheric current pattern response to southward and northward IMF turnings with dynamical supermag correlation networks. *Journal of Geophysical Research: Space Physics*, *122*, 1883– 1902. https://doi.org/10.1002/2016JA023686

Dods, J., Chapman, S. C., & Gjerloev, J. W. (2015). Network analysis of geomagnetic substorms using the supermag database of ground-based magnetometer stations. *Journal of Geophysical Research: Space Physics*, *120*(9), 7774–7784. https://doi.org/10.1002/2015JA021456

Donges, J. F., Zou, Y., Marwan, N., & Kurths, J. (2009). The backbone of the climate network [Article]. *Europhysics Letters*, *87*(4). https://doi.org/10.1209/0295-5075/87/48007

EPRI. (2017). *Space weather activity and the impact on the transmission system observed in September 2017 (Technical Report No. 3002011973)*. Electric Power Research Institute.

EPRI. (2018). *Node installation*. Supplemental Project Notice.

EPRI. (2018). *Sunburst network membership*. Supplemental Project Notice.

EPRI. (2020). *Improving conductivity models for geomagnetically induced current (GIC) estimation (Technical Report No. 3002017897)*. Electric Power Research Institute.

Gilman, D. L., Fuglister, F. J., & Mitchell, J. M. (1963). On the power spectrum of "red noise". *Journal of the Atmospheric Sciences*, *20*(2), 182–184. https://doi.org/10.1175/1520-0469(1963)020<0182:otpson>2.0.co;2

Gjerloev, J. W. (2009). A global ground-based magnetometer initiative. *Eos, Transactions-American Geophysical Union*, *90*(27), 230–231. https://doi.org/10.1029/2009EO270002

Gjerloev, J. W. (2012). The supermag data processing technique. *Journal of Geophysical Research: Space Physics*, *117*(A9), A09213. https://doi.org/10.1029/2012JA017683

Keiling, A., Angelopoulos, V., Runov, A., Weygand, J., Apatenkov, S. V., Mende, S., et al. (2009). Substorm current wedge driven by plasma flow vortices: Themis observations. *Journal of Geophysical Research: Space Physics*, *114*(A1), A00C22. https://doi.org/10.1029/2009JA014114

Kellerman, A. C., Mcgranaghan, R., Bortnik, J., Carter, B. A., Hughes, J., Arritt, R. F., et al. (2021). Geomagnetically induced currents at middle latitudes: 1. Quiet-time variability. *Space Weather*, *19*, e2021SW002729. https://doi.org/10.1029/2021SW002729

Lanzerotti, L. J. (2001).*Space weather effects on technologies*. In D. Song, H. J. Singer, & G. L. S. SiscoeWeather (Eds.), (Vol. 125, pp. 11–22). AGU Geophysical Monograph.

Lesher, R. L., Porter, J. W., & Byerly, R. T. (1994). Sunburst-A network of GIC monitoring systems. *IEEE Transactions on Power Delivery*, *9*(1), 128–137. https://doi.org/10.1109/61.277687

Lui, A. T. Y., & Najmi, A. H. (1997). Time-frequency decomposition of signals in a current disruption event. *Geophysical Research Letters*, *24*(24), 3157–3160. https://doi.org/10.1029/97gl03229

Malik, N., Bookhagen, B., Marwan, N., & Kurths, J. (2011). Analysis of spatial and temporal extreme monsoonal rainfall over south Asia using complex networks. *Climate Dynamics*, *39*(3–4), 971–987. https://doi.org/10.1007/s00382-011-1156-4

McGranaghan, R. M., Bhatt, A., Matsuo, T., Mannucci, A. J., Semeter, J. L., & Datta-Barua, S. (2017). Ushering in a new frontier in geospace through data science. *Journal of Geophysical Research: Space Physics*, *122*(12), 12586–12590. https://doi.org/10.1002/2017JA024835

Milgram, S. (1967). The small-world problem. *Psychology Today*, *1*(1), 60–67. https://doi.org/10.1037/e400002009-005

Newell, P. T., & Gjerloev, J. W. (2012). Supermag-based partial ring current indices. *Journal of Geophysical Research: Space Physics*, *117*(A5), A05215. https://doi.org/10.1029/2012JA017586

Newell, P. T., & Gjerloev, J. W. (2014). Local geomagnetic indices and the prediction of auroral power. *Journal of Geophysical Research: Space Physics*, *119*(12), 9790–9803. https://doi.org/10.1002/2014JA020524

Ngwira, C. M., Pulkkinen, A., Wilder, F. D., & Crowley, G. (2013). Extended study of extreme geoelectric field event scenarios for geomagnetically induced current applications. *AGU Space Weather*, *11*, 121–131. https://doi.org/10.1002/swe.20021

Nose, M., Iyemori, T., Takeda, M., Kamei, T., Milling, D., Orr, D., & Sumitomo, N. (1998). Automated detection of pi 2 pulsations using wavelet analysis: 1. Method and an application for substorm monitoring. *Earth Planets and Space*, *50*(9). https://doi.org/10.1186/bf03352169

Ochadlick, A. R., Jr, Kritikos, H. N., & Giegengack, R. (1993). Variations in the period of the sunspot cycle. *Geophysical Research Letters*, *20*(14), 1471–1474. https://doi.org/10.1029/93GL01593

Orr, L., Chapman, S., Gjerloev, J., & Guo, W. (2021). Network community structure of substorms using supermag magnetometers. *Nature Communications*, *12*(1). https://doi.org/10.1038/s41467-021-22112-4

Orr, L., Chapman, S. C., & Beggan, C. D. (2021). Wavelet and network analysis of magnetic field variation and geomagnetically induced currents during large storms. *AGU Space Weather*, *19*. e2021SW002772. https://doi.org/10.1029/2021SW002772

Orr, L., Chapman, S. C., & Gjerloev, J. W. (2019). Directed network of substorms using supermag ground-based magnetometer data. *Geophysical Research Letters*, *46*, 6268–6278. https://doi.org/10.1029/2019GL082824

Parkinson, W. D. (1959). Directions of rapid geomagnetic fluctuations. *Geophysical Journal of the Royal Astronomical Society*, *2*(1). https://doi.org/10.1111/j.1365-246x.1959.tb05776.x

Parkinson, W. D. (1962). The influence of continents and oceans on geomagnetic variations. *Geophysical Journal International*, *6*. https://doi.org/10.1111/j.1365-246x.1962.tb02992.x

Pirjola, R. (2000). Geomagnetically induced currents during magnetic storms. *IEEE Transactions on Plasma Science*, *28*(6), 1867–1873. https://doi.org/10.1109/27.902215

Pirjola, R. (2002). Review on the calculation of the surface electric and magnetic fields and geomagnetically induced currents in ground based technological systems. *Surveys in Geophysics*, *23*, 71–90. https://doi.org/10.1023/a:1014816009303

Pulkkinen, A., Bernabeu, E., Thomson, A., Viljanen, A., Pirjola, R., Boteler, D., et al. (2017). Geomagnetically induced currents: Science, engineering, and applications readiness. *Space Weather*, *15*(7), 828–856. https://doi.org/10.1002/2016SW001501

Schrijver, C. J., Kauristie, K., Aylward, A. D., Denardini, C. M., Gibson, S. E., Glover, A., et al. (2015). Understanding space weather to shield society: A global road map for 2015–2025 commissioned by COSPAR and ILWS. *Advances in Space Research*, *55*(12), 2745–2807. https://doi.org/10.1016/j.asr.2015.03.023

Smith, A., Rae, I., Forsyth, C., Watt, C., Murphy, K., & Mann, I. (2020). Diagnosing the time-dependent nature of magnetosphere-ionosphere coupling via ulf waves at substorm onset. *Journal of Geophysical Research: Space Physics*, *125*(11), e2020JA028573. https://doi.org/10.1029/2020ja028573

15427390, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021SW002727, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Steinhaeuser, K., Ganguly, A. R., & Chawla, N. V. (2011). Multivariate and multiscale dependence in the global climate system revealed through complex networks. *Climate Dynamics*, *39*(3–4), 889–895. https://doi.org/10.1007/s00382-011-1135-9

Torrence, C., & Compo, G. (1998). A practical guide to wavelet analysis. *Bulletin of the American Meterological Society*, *79*(1). https://doi.org/10.1175/1520-0477(1998)079<0061:apgtwa>2.0.co;2

Tsonis, A. A., Swanson, K. L., & Roebber, P. J. (2006). What do networks have to do with climate? *Bulletin of the American Meteorological Society*, *87*(5), 585–596. https://doi.org/10.1175/BAMS-87-5-585

Viljanen, A., & Pirjola, R. (1994). Geomagnetically induced currents in the Finnish high-voltage power system. *Surveys in Geophysics*, *15*, 383–408. https://doi.org/10.1007/bf00665999

Woodroffe, J. R., Morley, S. K., Jordanova, V. K., Henderson, M. G., Cowee, M. M., & Gjerloev, J. G. (2016). The latitudinal variation of geoelectromagnetic disturbances during large (Dst <-100 nt) geomagnetic storms. *Space Weather*, *14*(9), 668–681. https://doi.org/10.1002/2016SW001376

Yamazaki, Y., & Maute, A. (2017). Sq and EEJ—A review on the daily variation of the geomagnetic field caused by ionospheric dynamo currents. *Space Science Reviews*, *206*. https://doi.org/10.1007/s11214-016-0282-z

15427390, 2022, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021SW002727, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License